

$$V = I \cdot R$$

$$I = \frac{V}{R}$$

الدائرة المكافئة تستبدل الدايود ونقطته ببله مقاومة ← يستعمل نفس الراور

equivalent model اليوم التاريخ / /

منوان الدرس

1 Ideal mode:

forward: $V_D = 0$ $R = 0$ → short circuit

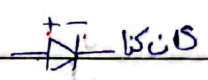
I_{Dmax} Depends on the circuit



reverse: $I = 0$ $R = \infty$

V_D Depends on the circuit

2 Simplified model:



forward:

تم تبديل الدايود ببطارية
عشان يكون مقاومة يعني $R \neq 0$

reverse:

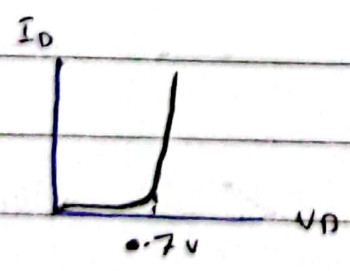
$R = 0$ مغل

3 piece - wise Linear model:



forward:

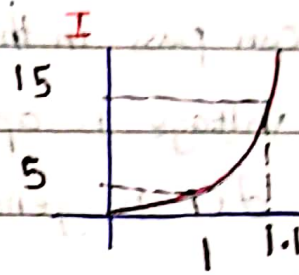
تم تبديل الدايود ببطارية ومقاومة



V-I characteristics

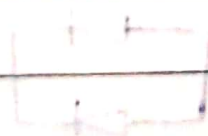
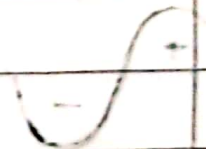
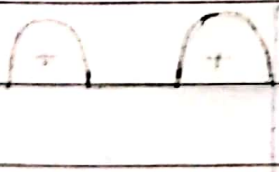
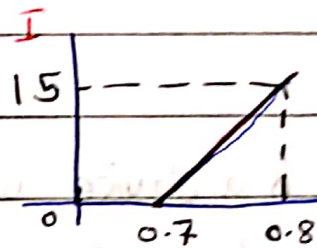
الواجب

$$R = \frac{V}{I} = \frac{1.1 - 1}{15 - 5} = \frac{0.1}{10} = 0.01 \text{ k}\Omega = 10 \Omega$$



$$R = \frac{0.8 - 0.7}{15 - 0} = \frac{0.1}{15}$$

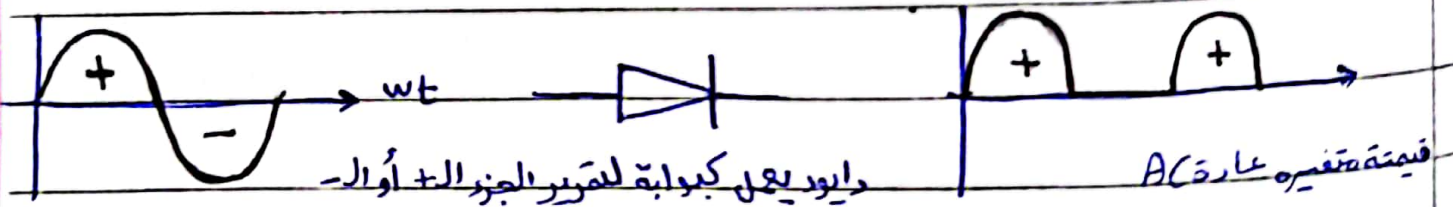
$$= 0.0067 \text{ k}\Omega = 6.7 \Omega$$



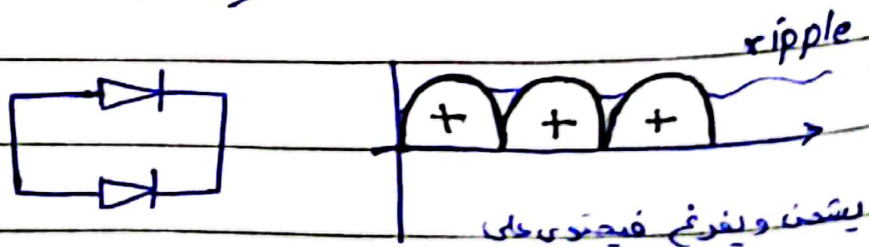
الواجب : الامثلة والتقاين بالانظمة

The pn junction diode conducts current in only one direction, and no current can pass in the other direction. Thus, when the alternating voltage is applied to pn junction diode the pulsating d.c. is generated across the resistance which is connected in series with the diode.

Rectifier: is a device which converts A.C. voltage into pulsating D.C. voltage.



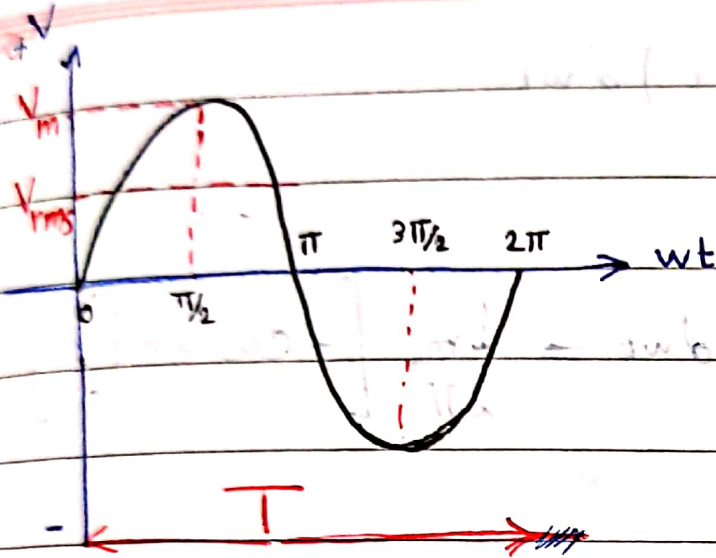
دايود يعمل كإيقاف للتيار في الاتجاه السالب أو العكس
في هذه الاتجاه يمر التيار الموجب فقط



يشحن ويفرغ في كل دورة

ripple is the amount of A.C. content in the output d.c. الخارج في الخارج AC في الخارج
واستخدم الكابتر لنقل التيار (لا يمكن إزالته) should be minimin can not remove

ليصبح قريب من ال DC غير نقي not pure



V_m → maximum voltage

القيمة العظمى

V_{rms} → root mean square voltage

القيمة الفعالة للتيار المتردد بين 0 و $\frac{\pi}{2}$

T → periodic Time (s)

f → frequency (Hz) قوانين تحويل القيمة الفعالة إلى عظمى والعكس

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

→

$$V_m = \sqrt{2} \cdot V_{rms}$$

Half Wave Rectifier

اليوم / / التاريخ

مقرن الدرس

Average

$$I_{DC} = \frac{1}{2\pi} \int_0^{2\pi} (I_m \sin \omega t) d\omega t$$

$-(1-1)$
 $-(\cos \pi - \cos 0)$

$$= \frac{1}{2\pi} \int_0^{\pi} (I_m \sin \omega t) d\omega t = \frac{I_m}{2\pi} \left[-\cos \omega t \right]_0^{\pi}$$

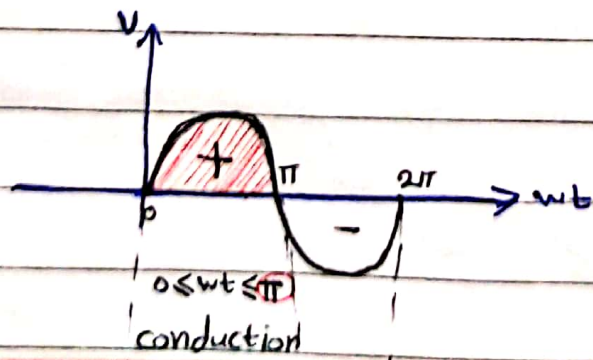
$$I_{DC} = \frac{I_m}{\pi}$$

$$I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} (I_m \sin \omega t)^2 d\omega t}$$

$$= \sqrt{\frac{I_m^2}{2\pi} \int_0^{\pi} (\sin^2 \omega t) d\omega t} = \sqrt{\frac{I_m^2}{2\pi} \int_0^{\pi} \frac{1}{2} (1 - \cos 2\omega t) d\omega t}$$

بقى الـ ωt

$$I_{rms} = \sqrt{\frac{I_m^2}{4\pi}} \rightarrow I_{rms} = \frac{I_m}{2}$$



التي يتكون منها الدايود

الواجب :

$$I_{DC} = \frac{1}{2\pi} \int_0^{2\pi} (I_m \sin \omega t) d\omega t$$

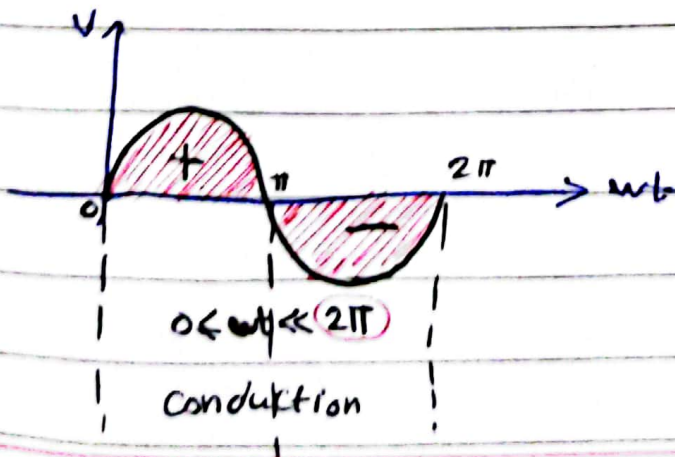
$$= \frac{1}{2\pi} 2 \int_0^{\pi} (I_m \sin \omega t) d\omega t = \frac{I_m}{\pi} \left[-\cos \omega t \right]_0^{\pi}$$

$$I_{DC} = 2 \frac{I_m}{\pi}$$

$$I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} (I_m \sin \omega t)^2 d\omega t}$$

$$= \sqrt{\frac{I_m^2}{2\pi} 2 \int_0^{\pi} (\sin^2 \omega t) d\omega t} = \sqrt{\frac{I_m^2}{\pi} \int_0^{\pi} \frac{1}{2} (1 - \cos 2\omega t) d\omega t}$$

$$I_{rms} = \sqrt{\frac{I_m^2}{2\pi} \pi} \rightarrow I_{rms} = \frac{I_m}{\sqrt{2}}$$



عند $r_f \neq 0$

$$I_m = \frac{V_m}{r_f + R_L}$$

H.W.

$$I_{DC} = \frac{I_m}{\pi}$$

$$V_{DC} = I_{DC} * R_L$$

F.W

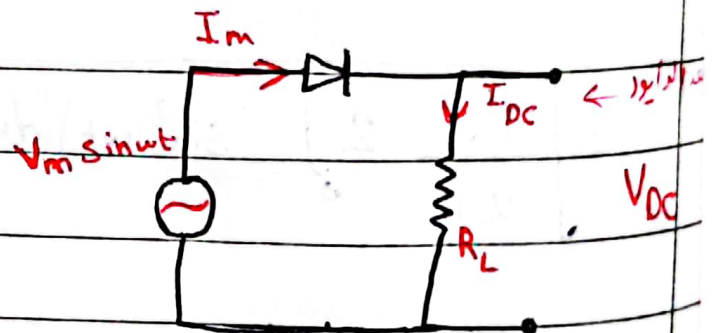
$$I_{DC} = \frac{2 I_m}{\pi}$$

$$V_{DC} = I_{DC} * R_L$$

$r_f = 0$ ideal diode

$$I_m = \frac{V_m}{R_L}$$

عند الدايود



$$V_{DC} = I_{DC} * R_L$$

عند التوازي مع الدايود

$$V_{DC} = \frac{I_m}{\pi} * R_L$$

$$V_{DC} = \frac{\frac{V_m}{R_L}}{\pi} * R_L$$

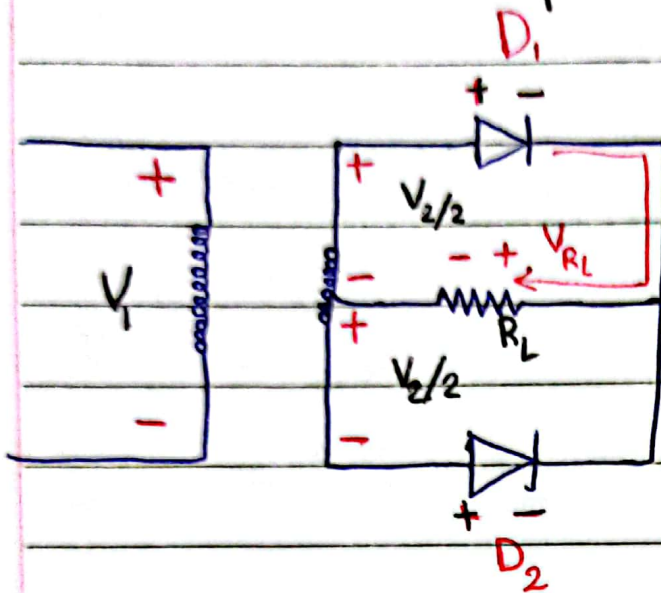
$$V_{DC} = \frac{V_m}{\pi}$$

when $r_f = 0$

الواجب

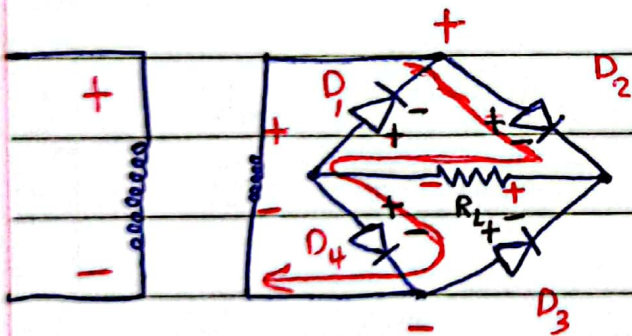
III Center tap:

for positive half cycle



D_1 is on

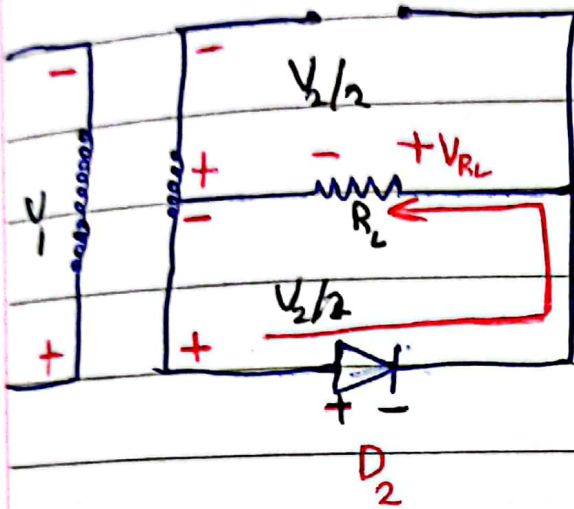
D_2 is off



D_2 and D_4 are both on

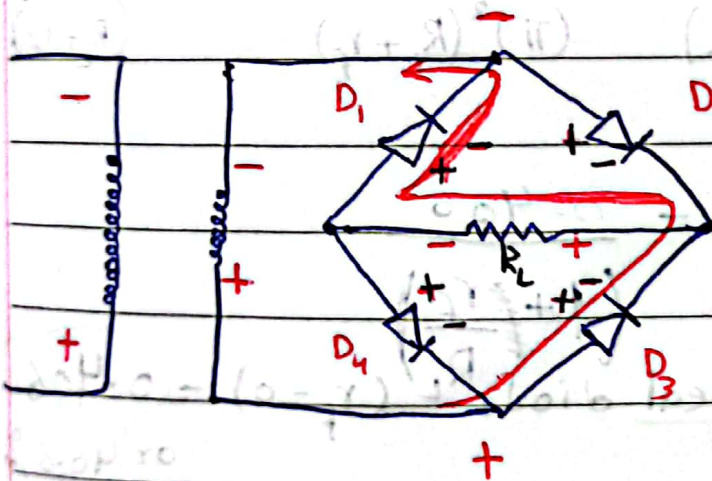
D_1 and D_3 are both off

for negative half cycle



D_1 is off

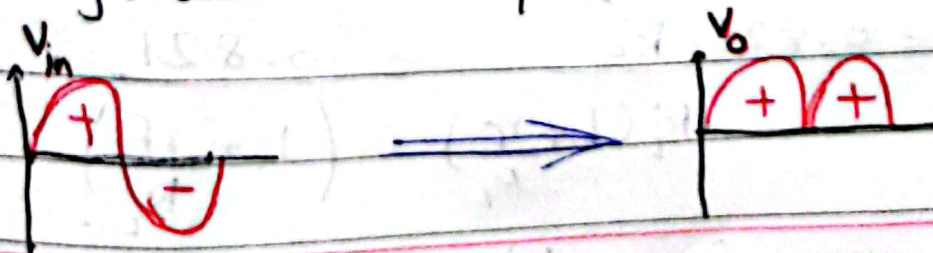
D_2 is on



D_1 and D_3 are both on

D_2 and D_4 are both off

The current through Load is the same for both half cycle of the input.



الواجب :

Efficiency is ratio of the dc output power to the ac input power.

$$\eta = \frac{P_{dc}}{P_{ac}}$$

$$P_{dc} = I_{dc}^2 \times (R_L)$$

$$P_{ac} = I_{rms}^2 \times (R_L + r_f)$$

Half wave: $\eta = \frac{\left(\frac{I_m}{\pi}\right)^2 R_L}{\left(\frac{I_m}{2}\right)^2 (R_L + r_f)} = \frac{(2)^2 R_L}{(\pi)^2 (R_L + r_f)} = 0.406 \frac{R_L}{(R_L + r_f)}$

$$= 0.406 \frac{R_L}{R_L (1 + \frac{r_f}{R_L})} = 0.406 \frac{1}{1 + \frac{r_f}{R_L}}$$

maximum efficiency for an ideal diode ($r_f = 0$) = 0.406 or 40.6%

Full wave: $\eta = \frac{\left(\frac{2I_m}{\pi}\right)^2 R_L}{\left(\frac{I_m}{\sqrt{2}}\right)^2 (R_L + r_f)} = \frac{(2\sqrt{2})^2 R_L}{(\pi)^2 (R_L + r_f)} = 0.821 \frac{R_L}{(R_L + r_f)}$

$$= 0.821 \frac{R_L}{R_L (1 + \frac{r_f}{R_L})} = 0.821 \frac{1}{1 + \frac{r_f}{R_L}}$$

for an ideal diode = ($r_f = 0$) = 0.821 or 81.2%

* Ripple factor The amount of AC in the output of the rectifier.

* Rectifier output is not pure DC.

* Ripple factor tells us how smooth the output.

* How can we minimize the ripple?

we can minimize the ripple by using capacitor in parallel with Load resistance.

الواجب، الخشتقاق هم جداً في الملائمة

(SUPER MAAS)

Transistor has three terminals are called:

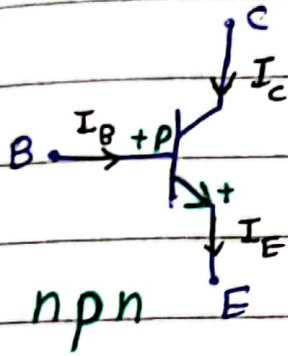
① base.

applications: ① Amplifier,

② emitter.

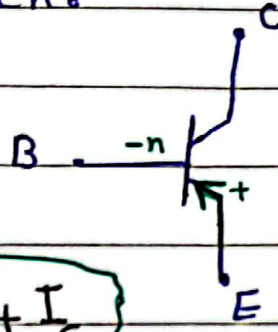
② Switch.

③ collector.



$$I_E = I_B + I_C$$

input output



لماذا

pnp

أقل استخداماً؟

The arrow indicate to the direction of holes

what the different between switch in diode and transistor?

transistor have high speed switch and Low delay than diode.

Three regions:

input = output = 0

- ① cut off → off switch , $I_B = I_C = 0$ التيار جف
- ② active → Amplifier I_{sat} يستغل فالتيارين هـ
- ③ Saturation → on switch , $I_C = I_{sat}$ التيار الكلي ما يمكن

الترانزستور يوجد في اجهزة الاستقبال لتكبير ال signal أو ال Voltage بعد القوس.

الواجب :